



```
In [1]: !pip install tensorflow
```

Requirement already satisfied: tensorflow in c:\users\mandi\anaconda3\lib\site-packages (2.21.0)

Requirement already satisfied: absl-py>=1.0.0 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (2.4.0)

Requirement already satisfied: astunparse>=1.6.0 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (1.6.3)

Requirement already satisfied: flatbuffers>=25.9.23 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (25.12.19)

Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (0.7.0)

Requirement already satisfied: google_pasta>=0.1.1 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (0.2.0)

Requirement already satisfied: libclang>=13.0.0 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (18.1.1)

Requirement already satisfied: opt_einsum>=2.3.2 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (3.4.0)

Requirement already satisfied: packaging in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (25.0)

Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (7.34.0)

Requirement already satisfied: requests<3,>=2.21.0 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (2.32.5)

Requirement already satisfied: setuptools in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (80.9.0)

Requirement already satisfied: six>=1.12.0 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (1.17.0)

Requirement already satisfied: termcolor>=1.1.0 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (3.3.0)

Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (4.15.0)

Requirement already satisfied: wrapt>=1.11.0 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (1.17.0)

Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (1.78.0)

Requirement already satisfied: keras>=3.12.0 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (3.13.2)

Requirement already satisfied: numpy>=1.26.0 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (2.3.5)

Requirement already satisfied: h5py<3.15.0,>=3.11.0 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (3.14.0)

Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (0.5.4)

Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\mandi\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in c:\users\mandi\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\mandi\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\mandi\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2026.1.4)

Requirement already satisfied: wheel<1.0,>=0.23.0 in c:\users\mandi\anaconda3\lib\site-packages (from astunparse>=1.6.0->tensorflow) (0.45.1)

Requirement already satisfied: rich in c:\users\mandi\anaconda3\lib\site-packages (from keras>=3.12.0->tensorflow) (14.2.0)

Requirement already satisfied: namex in c:\users\mandi\anaconda3\lib\site-packages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in c:\users\mandi\anaconda3\lib\site-packages (from keras>=3.12.0->tensorflow) (0.19.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\mandi\anaconda3\lib\site-packages (from rich->keras>=3.12.0->tensorflow) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\mandi\anaconda3\lib\site-packages (from rich->keras>=3.12.0->tensorflow) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in c:\users\mandi\anaconda3\lib\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.12.0->tensorflow) (0.1.2)

```
In [2]: import pandas as pd
import numpy as np
```

```
In [3]: train_df = pd.read_csv("Documents/train.csv")
test_df = pd.read_csv("Documents/test.csv")

print("Train shape:", train_df.shape)
print("Test shape:", test_df.shape)
```

Train shape: (42000, 785)
Test shape: (28000, 784)

```
In [4]: X_train = train_df.drop("label", axis=1)
y_train = train_df["label"]

X_test = test_df

print("X_train:", X_train.shape)
print("y_train:", y_train.shape)
print("X_test:", X_test.shape)
```

X_train: (42000, 784)
y_train: (42000,)
X_test: (28000, 784)

```
In [5]: X_train = X_train / 255.0
X_test = X_test / 255.0
```

```
In [6]: import numpy as np

X_train = np.array(X_train)
y_train = np.array(y_train)
X_test = np.array(X_test)
```

```
In [7]: from sklearn.model_selection import train_test_split

X_train, X_val, y_train, y_val = train_test_split(
    X_train, y_train,
    test_size=0.1,
    random_state=42
)

print("Training:", X_train.shape)
```

```
print("Validation:", X_val.shape)
```

Training: (37800, 784)

Validation: (4200, 784)

```
In [8]: import tensorflow as tf
        from tensorflow import keras

        model = keras.Sequential([
            keras.Input(shape=(784,)),
            keras.layers.Dense(256, activation='relu'),
            keras.layers.Dropout(0.3),
            keras.layers.Dense(128, activation='relu'),
            keras.layers.Dense(10, activation='softmax')
        ])

        model.compile(
            optimizer='adam',
            loss='sparse_categorical_crossentropy',
            metrics=['accuracy']
        )

        history = model.fit(
            X_train, y_train,
            epochs=15,
            batch_size=64,
            validation_data=(X_val, y_val)
        )
```

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

Epoch 1/15

591/591  **9s** 12ms/step - accuracy: 0.9001 - loss: 0.3332 - val_accuracy: 0.9502 - val_loss: 0.1576

Epoch 2/15

591/591  **7s** 11ms/step - accuracy: 0.9561 - loss: 0.1437 - val_accuracy: 0.9633 - val_loss: 0.1169

Epoch 3/15

591/591  **10s** 11ms/step - accuracy: 0.9652 - loss: 0.1093 - val_accuracy: 0.9652 - val_loss: 0.1096

Epoch 4/15

591/591  **6s** 11ms/step - accuracy: 0.9724 - loss: 0.0886 - val_accuracy: 0.9724 - val_loss: 0.0888

Epoch 5/15

591/591  **5s** 9ms/step - accuracy: 0.9773 - loss: 0.0724 - val_accuracy: 0.9719 - val_loss: 0.0899

Epoch 6/15

591/591  **7s** 12ms/step - accuracy: 0.9796 - loss: 0.0619 - val_accuracy: 0.9762 - val_loss: 0.0829

Epoch 7/15

591/591  **6s** 10ms/step - accuracy: 0.9815 - loss: 0.0564 - val_accuracy: 0.9764 - val_loss: 0.0806

Epoch 8/15

591/591  **9s** 9ms/step - accuracy: 0.9821 - loss: 0.0548 - val_accuracy: 0.9729 - val_loss: 0.0869

Epoch 9/15

591/591  **7s** 12ms/step - accuracy: 0.9849 - loss: 0.0449 - val_accuracy: 0.9771 - val_loss: 0.0832

Epoch 10/15

591/591  **7s** 11ms/step - accuracy: 0.9853 - loss: 0.0431 - val_accuracy: 0.9760 - val_loss: 0.0838

Epoch 11/15

591/591  **11s** 12ms/step - accuracy: 0.9872 - loss: 0.0384 - val_accuracy: 0.9767 - val_loss: 0.0916

Epoch 12/15

591/591  **8s** 9ms/step - accuracy: 0.9874 - loss: 0.0389 - val_accuracy: 0.9783 - val_loss: 0.0831

Epoch 13/15

591/591  **8s** 13ms/step - accuracy: 0.9890 - loss: 0.0318 - val_accuracy: 0.9786 - val_loss: 0.0749

Epoch 14/15

591/591  **8s** 9ms/step - accuracy: 0.9885 - loss: 0.0340 - val_accuracy: 0.9783 - val_loss: 0.0848

Epoch 15/15

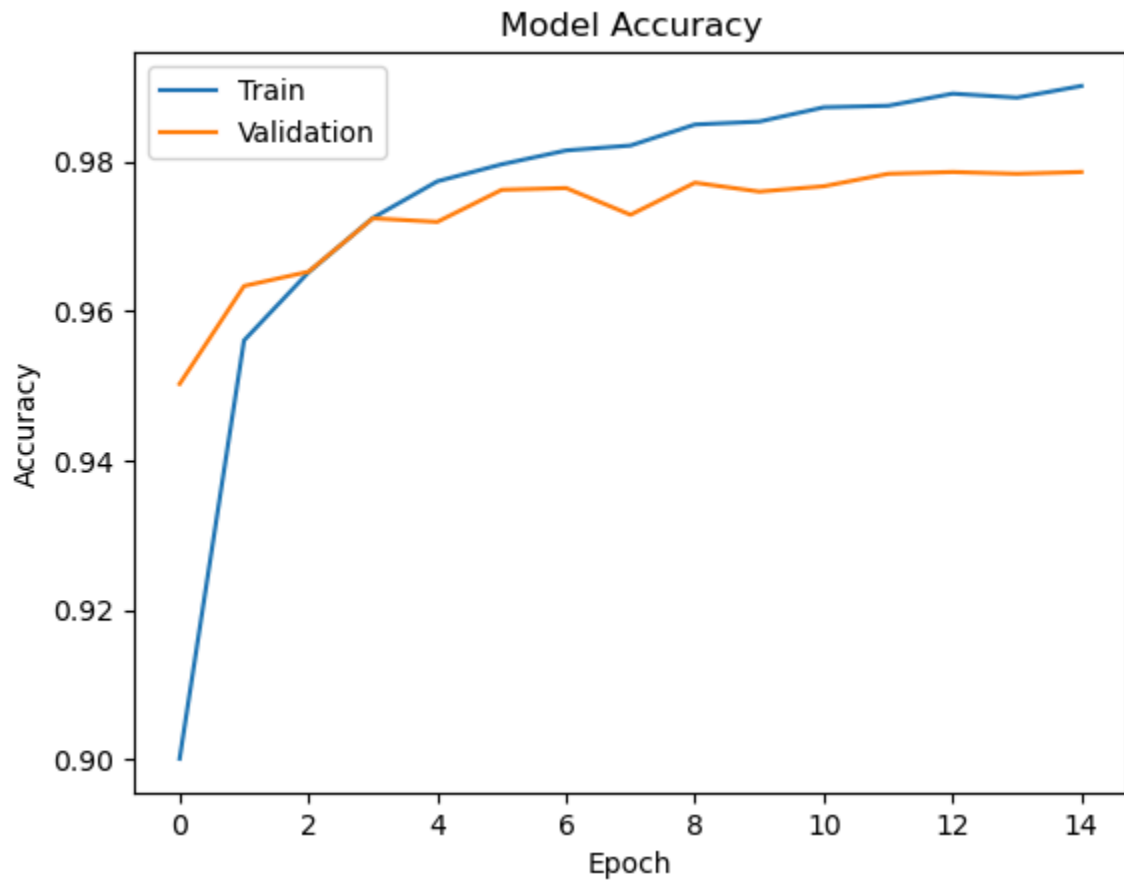
591/591  **11s** 11ms/step - accuracy: 0.9901 - loss: 0.0292 - val_accuracy: 0.9786 - val_loss: 0.0847

```
In [9]: val_loss, val_acc = model.evaluate(X_val, y_val)
        print("Validation Accuracy:", val_acc)
```

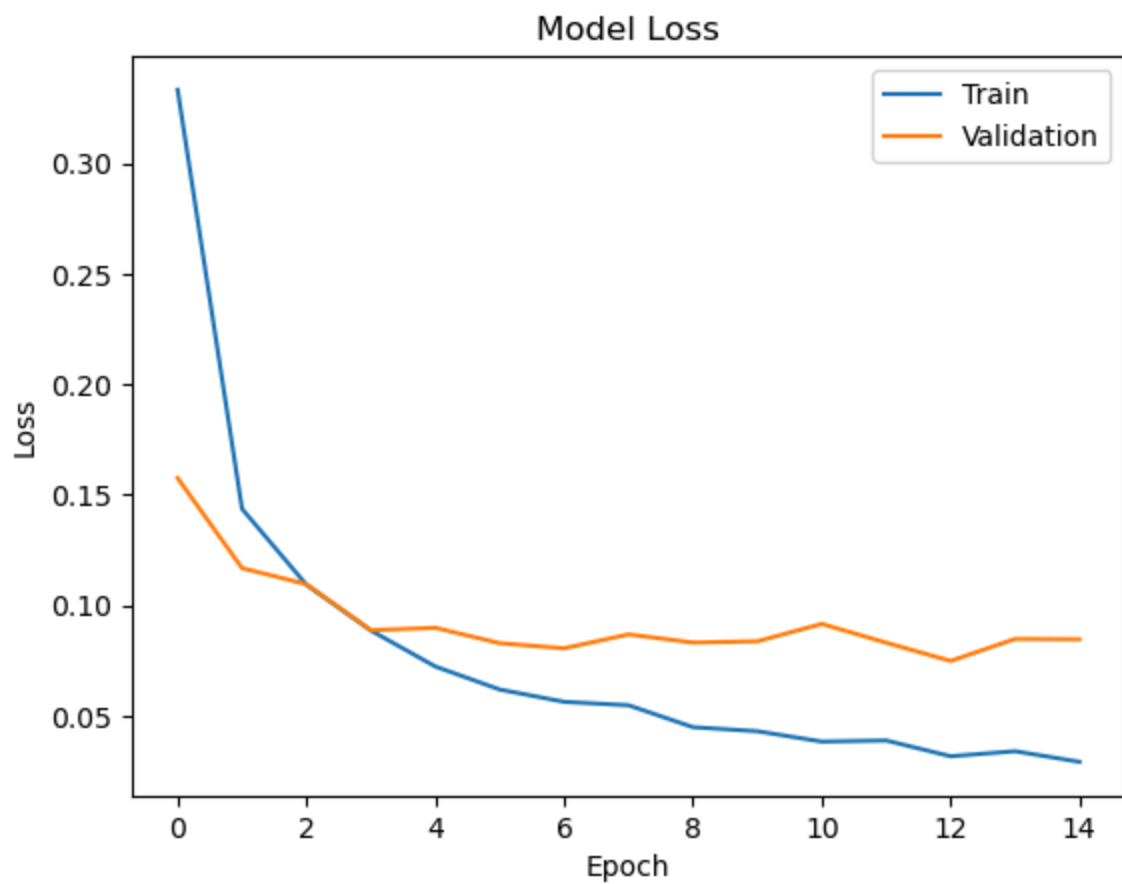
132/132  **1s** 5ms/step - accuracy: 0.9786 - loss: 0.0847
Validation Accuracy: 0.9785714149475098

```
In [10]: import matplotlib.pyplot as plt
```

```
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title('Model Accuracy')
plt.ylabel('Accuracy')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'])
plt.show()
```



```
In [11]: plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('Model Loss')
plt.ylabel('Loss')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'])
plt.show()
```



```
In [12]: predictions = model.predict(X_test)
         predicted_labels = predictions.argmax(axis=1)
```

875/875 ————— 2s 3ms/step

```
In [13]: submission = pd.DataFrame({
         "ImageId": range(1, len(predicted_labels) + 1),
         "Label": predicted_labels
       })

         submission.to_csv("submission.csv", index=False)
```